



STUDY BUDDY FINDER

CS3354 Software
Engineering

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Project Objective



Objective

Study Buddy Finder is a web application designed to help students connect with classmates for academic collaboration by finding compatible study partners and organizing study groups.



Main Goals

- Help students find study partners based on shared courses.
- Allow students to create and join study groups.
- Schedule study sessions online or in person.
- Improve collaboration and academic success.
- Provide a simple, secure, and user-friendly platform.

Cost Estimation



Estimation Method

- Function Point Analysis (FP)



Development Effort

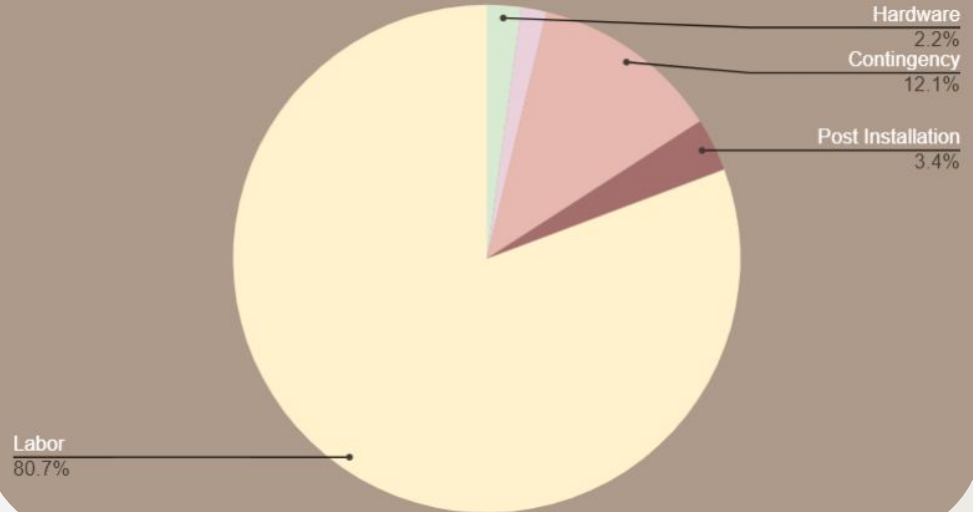
- 5 Team Members
- 675 Total Team Hours
- 135 Hours per Member



Estimated Costs

- **Labor Cost:** \$20,250
- **Hardware Cost:** \$550
- **Software Cost:** \$0
- **Personnel Cost:** \$21,500
- **Overall Project Cost:** \$25,087.50

Project Cost Breakdown





Project Timeline



Phase	Duration	Dates
Requirements Analysis	1 Week	June 14 – June 20
System Design	1 Week	June 21 – June 27
UI/UX Prototype	1 Week	June 28 – July 4
Backend Development	2 Weeks	July 5 – July 18
Frontend Development	2 Weeks	July 19 – August 1
Integration & Testing	1 Week	August 2 – August 8
Documentation & Final Presentation	5 Days	August 9 – August 13



Functional & Non-Functional Requirements



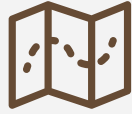
Functional Requirements - these describe what the system does.

- User registration and login
- Manage student profiles
- Search for study partners
- Create and join study groups
- Schedule study sessions



Non-Functional Requirements - these describe how the system should perform.

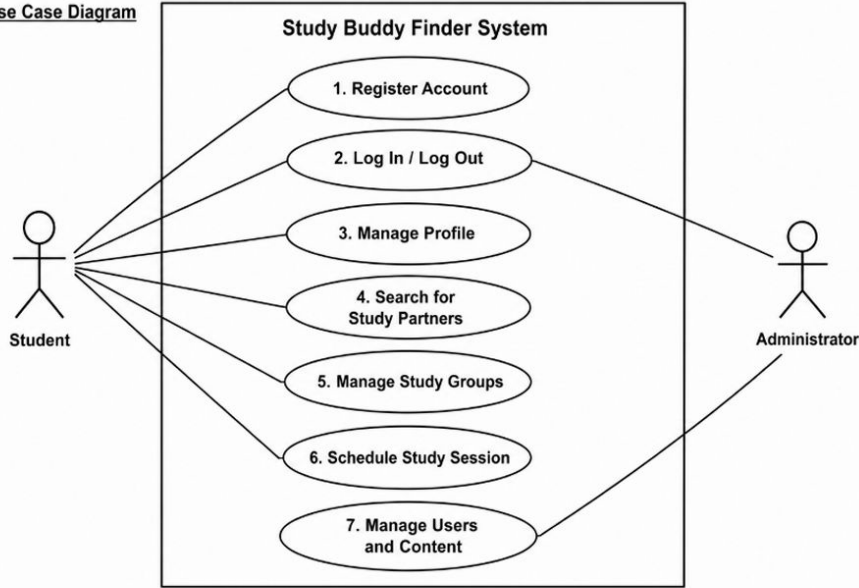
- Usability: Easy-to-use interface
- Performance: Search results load within 3 seconds
- Security: Passwords are encrypted and accounts are protected
- Reliability: System available 99% of the time
- Compatibility: Works on modern web browsers and mobile devices



Use Case Diagram



Use Case Diagram



Key Features

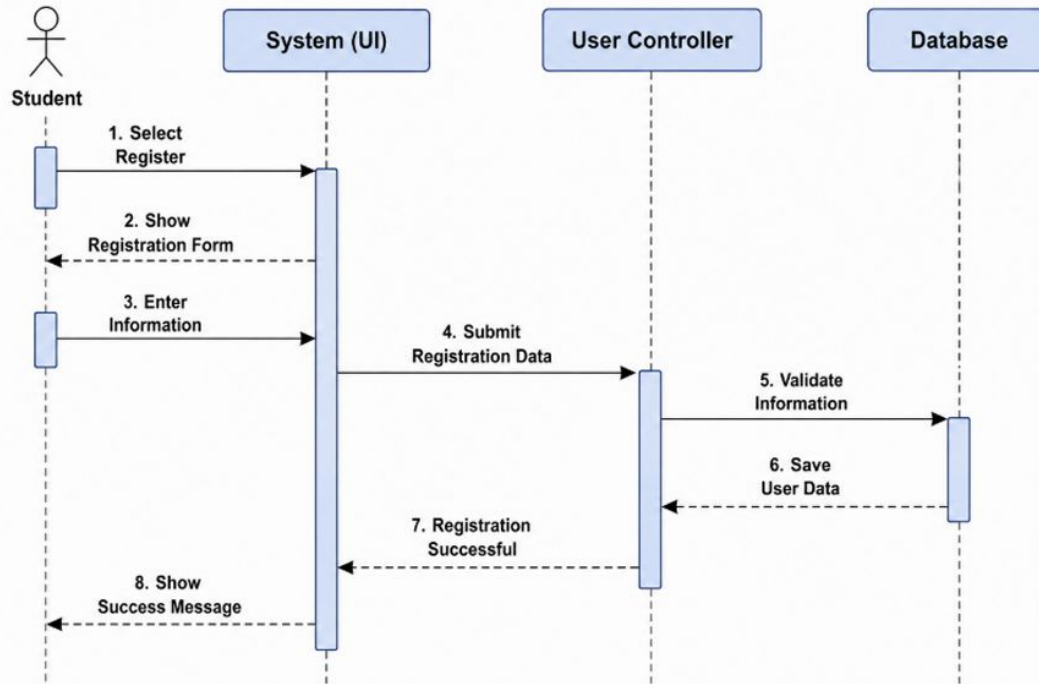
- Student registers and logs into the system.
- Students manage their profiles and study preferences.
- Students search for study partners with similar courses.
- Students create and join study groups.
- Students schedule study sessions.
- Administrators manage user accounts and system content.



Sequence Diagram



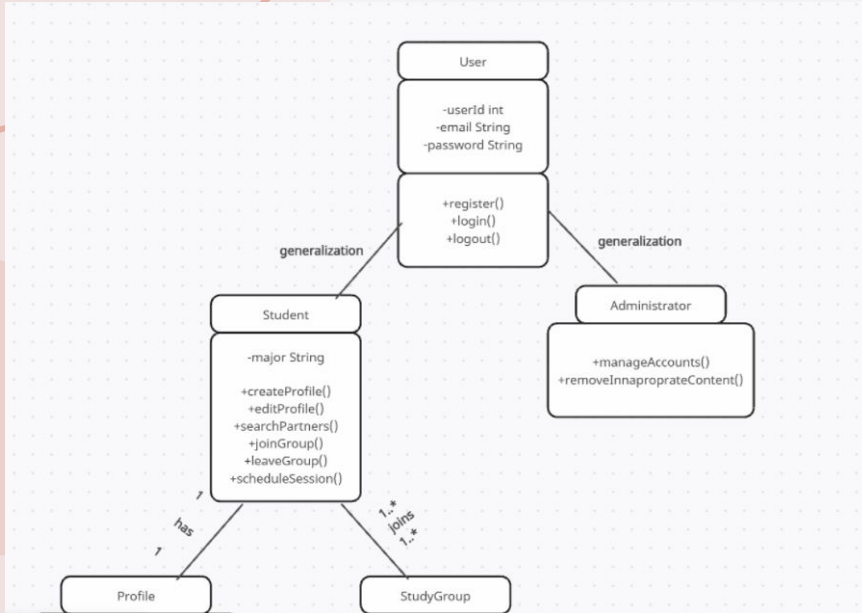
UC1: Register Account



- Registration Process
- Student selects Register.
- System displays the registration form.
- Student enters account information.
- Controller validates and saves the data.
- System confirms successful registration.

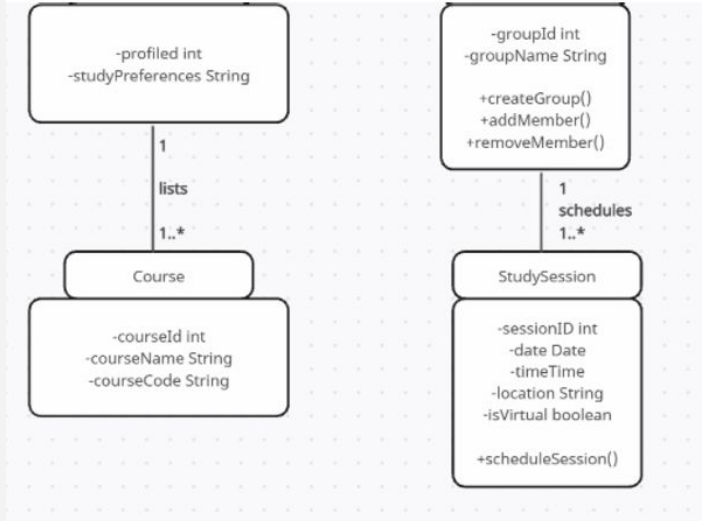


Class Diagram



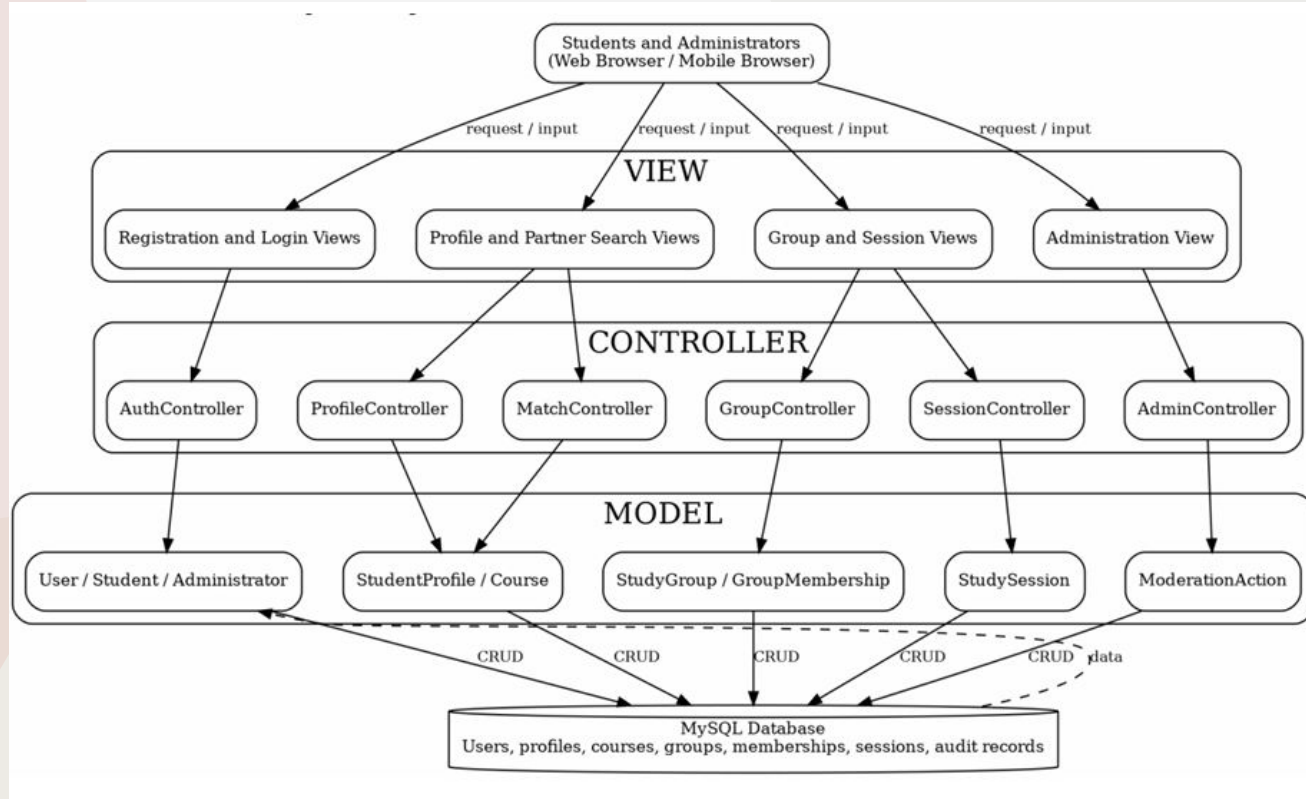
Main Classes

- Student – Stores student information and account details.
- Study Group – Represents study groups that students can create or join.
- Study Session – Stores information about scheduled study meetings.
- Administrator – Manages users and system content.
- Database – Stores user profiles, study groups, and session data.





Architectural Design – MVC Pattern





Architectural Design – MVC Pattern



→ Model

- Stores and manages application data
- Includes users, profiles, study groups, and study sessions
- Communicates with the database

→ View

- Displays the user interface
- Includes registration, profile, search, group, and scheduling screens
- Shows information to students and administrators

→ Controller

- Processes user actions
- Handles registration, login, search, group management, and scheduling
- Connects the View to the Model

→ Why MVC was selected

- Separates the interface, system logic, and data
- Makes the application easier to develop, test, and maintain
- Allows team members to work on different parts of the system independently



Comparison with Similar Applications



Feature	Study Buddy Finder	Meetup	Study Together
Find study partners	✓	✗	✓
Course-based matching	✓	✗	✗
Create study groups	✓	✓	✗
Schedule study sessions	✓	✓	✗
Student-focused	✓	✗	✓

- Designed specifically for students.
- Matches users based on shared courses and academic interests.
- Supports creating and joining study groups.
- Includes study session scheduling.
- More education-focused than general social networking apps.

Test Cases

Test Case	Input	Expected Result	Status
Login (Pass)	Valid email and password	User logs in successfully and is taken to the dashboard.	 Pass
Login (Fail)	Valid email with incorrect password	System displays an error message and prevents login.	 Fail

Purpose

- Verify that the system functions correctly.
- Ensure user inputs are validated.
- Confirm the application behaves as expected for both valid and invalid inputs.



UX Prototype

Study Buddy Finder

Welcome back

Continue to your private student workspace.

University email

rxr@utdallas.edu

Password

.....

Log in

[Create an account](#)

Study Buddy Finder

Create your account

Use your university identity while keeping your contact information private.

Display name

Raisa Reza

University email

rxr@utdallas.edu

Password

.....

10+ characters

Create account

[Already have an account? Log in](#)

Set up your study profile

A few private details help us rank compatible classmates without exposing your email or detailed schedule.

1 Profile 2 Courses 3 Availability 4 Matches

1. Profile

Display name

Diego

Major

Computer Science

Academic interests

Software testing, Exam preparation

Preferred study mode

In person

Online

Flexible

2. Active courses

CS 3354 — Software Engineering — Fall 2026

Active course • used for matching

CS 3345 — Data Structures — Fall 2026

Active course • used for matching

+ Add another course

3. Private availability

Detailed availability is private and only used for compatibility scoring.

Monday • 6:00 PM–8:00 PM

Private window

Tuesday • 6:00 PM–9:00 PM

Private window

Saturday • 10:00 AM–1:00 PM

Private window

+ Add availability window

See matches

Compatible study buddies

Ranked by shared courses, preferences, interests, and private availability overlap.

Barbara Kim

92% compatible

Computer Science • In person

Why this match

2 shared courses

Compatible evening availability

Shared interest in exam preparation

Jordan Kim

86% compatible

Software Engineering • Flexible

Why this match

CS 3354 in common

Compatible weekday availability

Prefers structured review sessions

Priya Patel

78% compatible

Computer Science • Online

Why this match

CS 3345 in common

Compatible weekend availability

Shared interest in problem solving

Dashboard

Matches

Groups

Join a Study Group

Course-based groups keep membership, sessions, and discussion inside approved spaces.

Create a group

CS 3354

CS 4347 Project - Team 5

Work on deliverable 3 together

View

Request to join

4 / 5 members

In person

CS 3345

Data Structures & Algorithms Midterm Prep

Practice complexity analysis and review trees

View

Request to join

3 / 6 members

Flexible

CS 3345

Data Structures Study Circle

Weekly problem-solving sessions for trees, graphs, and hashing.

View

Request to join

5 / 6 members

Online

Create a study group

Course

CS 3354 Software Engineering

Group name

Final Exam Prep

Description

Study for the final exam and share study resources

Capacity

5 students

Study mode

Online

Create group

Study Buddy Finder

STUDENT WORKSPACE

Dashboard

Matches

Groups

Privacy center

Log out

Coordinator tools

Final Exam Prep

Schedule session

Schedule a session

Start time

Tuesday, 6:00 PM

End time

Tuesday, 8:00 PM

Location or virtual link

ECSS 2.102

Agenda

Study for the final exam and share study resources

Schedule session

Moderation queue

Review reports without opening unrelated private student data.

Open

Message report

Reason: Personal attack in group discussion

Private email addresses and detailed availability are intentionally excluded.

Remove reported message

Under review

Account report

Reason: Repeated harassment across approved groups

Private email addresses and detailed availability are intentionally excluded.

Deactivate reported account



Challenges, Lessons Learned & Conclusion

Challenges

- Coordinating tasks and schedules among team members.
- Estimating project cost, effort, and timeline.
- Designing a user-friendly application that meets student needs.

Lessons Learned

- Improved teamwork and communication.
- Gained experience with software engineering practices and UML diagrams.
- Learned the importance of planning, testing, and user-centered design.

Conclusion

- Study Buddy Finder provides an organized way for students to connect and collaborate.
- The project demonstrates how software engineering principles can be applied to solve a real-world problem.
- Future enhancements could include messaging, notifications, and mobile app support.